

Scalable Multivariate NTT Multiplier

Extending a Mixed-Radix NTT Architecture

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Problem Statement

We compute:

$$C(X) = A(X) \cdot B(X) \bmod (X^N + 1)$$

Large NTTs suffer from:

- High memory usage
- Routing congestion
- Large twiddle ROMs
- Poor scalability for large N

Goal:

- Scale the existing mixed-radix NTT
- Reuse smaller NTT cores efficiently

Core Idea: Multivariate NTT

Instead of a 1-D polynomial:

$$[a_0, a_1, a_2, \dots, a_{N-1}]$$

reshape coefficients into a 2-D tensor:

$$\begin{bmatrix} a_0 & a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 & a_7 \\ a_8 & a_9 & a_{10} & a_{11} \end{bmatrix}$$

Using:

$$X_2 = X_1^{K/2}$$

This converts:

$$A(X) \rightarrow A(X_1, X_2)$$

Scaling the Original NTT Architecture

Traditional flow:

Large NTT $\rightarrow$ Multiply $\rightarrow$ Inverse NTT

Multivariate extension:

Tensor Mapping $\rightarrow$ Row NTTs $\rightarrow$ Transpose $\rightarrow$ Column NTTs
 $\rightarrow$ Pointwise Multiply $\rightarrow$ Inverse Column NTTs
 $\rightarrow$ Inverse Transpose $\rightarrow$ Inverse Row NTTs
 $\rightarrow$ 1D Reconstruction

Key idea:

- Reuse existing mixed-radix NTT cores
- Add tensor mapping + transpose network
- Replace one large NTT with many smaller NTTs

Multivariate NTT Architecture

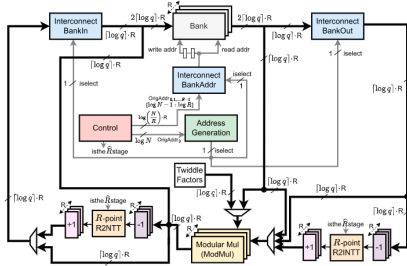


Fig. 6. An overall polynomial multiplication architecture based on high-radix/mixed-radix in-place NTT over fermat modulus. “-1” denotes the conversion from normal representation to D1 representation. “+1” denotes the conversion from D1 representation to normal representation.

Figure: Original NTT Multiplier Architecture

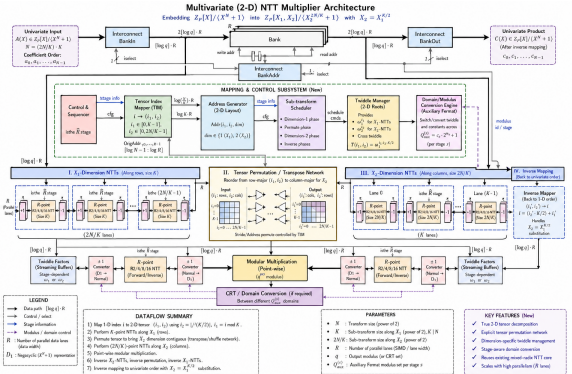


Figure: Multivariate NTT Multiplier Architecture

Architecture Operation

- 1 Map 1-D polynomial into a 2-D tensor
- 2 Perform row-wise small NTTs
- 3 Transpose tensor
- 4 Perform column-wise small NTTs
- 5 Pointwise modular multiplication
- 6 Inverse transforms + output reconstruction

Advantages:

- Better memory locality
- Easier routing
- Higher parallelism
- Scalable for large NTT sizes

Key Contributions:

- Scalable multivariate NTT architecture
- Reuse of existing mixed-radix NTT cores
- Tensor-based decomposition
- Improved FPGA/ASIC scalability

Applications:

- Fully Homomorphic Encryption (FHE)
- Post-Quantum Cryptography
- Large polynomial accelerators

Thank You